Earnings Surprise, Firm Valuation, and Market Efficiency: Evidence from China

Corporate Finance Team Project

Team Members: Yingqi Tan, Zhesheng Xu, Hao Yang

FIN803 Corporate Finance

Motivation

- Earnings-related disclosures update expectations about future cash flows and should affect valuation.
- The key issue is not whether a noisy long-window drift exists, but whether a cleaner design identifies defensible short-run pricing adjustment.
- Tushare Pro allows us to move from a weak internal proxy to a stronger event-expectation framework.

Research Question

Core Question

Do China A-share earnings-related events generate short-run abnormal return once surprise is measured relative to stricter sell-side expectations?

- Old version: guidance-only proxy design.
- New version: Tushare-first framework with explicit event types and pre-event expectation matching.
- Final framing emphasizes limited short-run valuation adjustment, not aggressive long-run inefficiency claims.

Tushare-first Research Framework

- Sell-side expectation module: `report_rc`
- Event modules: `forecast`/`forecast_vip`, `express`/`express_vip`, `fina_indicator`
- Market/control module: `daily_basic`, stock prices, market index
- Legacy guidance-only path is retained only as fallback / comparison

Why the Old Proxy Was Weak

- The older `ES_main`/`ES_std` path inferred expectations from prior guidance and historical heuristics.
- That is weaker than a true pre-event sell-side benchmark.
- It also mixes event types and makes look-ahead control harder to audit.
- So the old path now serves as legacy robustness rather than headline evidence.

Standardized Event Types

- `preannouncement`: management forecast / guidance event
- `revision`: updated forecast for the same period
- `express`: earnings express release
- `formal_release`: formal realized financial release
- Event trading date: first trading day after the disclosure date

Signal Construction

- Main expectation source: pre-event `report_rc`
- Forecast surprise: management forecast vs sell-side expectation
- Express surprise: express result vs sell-side expectation
- Final surprise: realized result vs sell-side expectation
- Main standardized signal: `main_surprise_std`

Key Principle

Do not silently collapse missing sell-side matches into the old proxy path; weak matches must be labeled and audited.

Controls and Filters

- Controls from `daily_basic`: total market value, float market value, turnover, PE, PB, PS
- Tradability filters: ST exclusion, minimum listing history, valid event-day return, positive event-day volume
- Event windows: `CAR3`, `CAR5`, `CAR10`, `CAR20`, `CAR60`

Preferred Regression

$$CAR_w = \alpha + \beta_1 main\_surprise\_std + \beta_2 \log(total\_mv) + \beta_3 beta + \beta_4 book\_to\_market + \beta_5 turnover20 +$$

- Standard errors clustered by firm.
- Headline window: `CAR5`.
- Additional windows are used for robustness and horizon comparison.

Current Test-Run Audit

- Current test mode uses 120 stocks.
- Raw pull succeeded for `forecast`, `express`, `fina_indicator`, and `report_rc`.
- Legacy guidance fallback produced a usable event panel.
- The strict Tushare-first sample is currently the bottleneck because coverage and filter alignment remain tight.

What Improved in the Codebase

- Added explicit Tushare-first config and directory structure.
- Added endpoint normalization, expectation alignment, event construction, and audit-output modules.
- Fixed the incomplete `daily_basic` field request so turnover and valuation controls are available.
- Added endpoint capability checks and run-level audit notes.

Why Coverage Is the Main Limitation

- The strict design requires pre-event `report_rc` matches for the same fiscal period.
- That improves identification but sharply reduces usable sample coverage.
- So the main research tradeoff is now sample purity versus sample size.
- This is preferable to overstating evidence from a weak expectation proxy.

Main Conclusion

- The repository is now organized around a stronger Tushare-first research design.
- The old weak-proxy path is preserved only as fallback / comparison.
- The most defensible headline remains: cleaner earnings-surprise measurement should be evaluated through short-run valuation adjustment, not strong long-run drift claims.
- The main empirical next step is improving matched-sample coverage while preserving strict pre-event alignment.

Final Takeaway

Recommended headline

A Tushare-first earnings-surprise design is more credible than the old guidance-only proxy, but the final inference should remain cautious until strict matched-sample coverage is stronger.